MCO-1D ORNITHOPTER

Bill of Materials — Revision 14

Complete parts list reflecting the fully corrected architecture: 8 independent motor-sector assemblies (no shared drive), 4 ODrive v3.6 boards (2 motors each), 2 MCP3208 ADCs (pitch and sweep feedback), and the complete sensing suite for full aerodynamic state knowledge per wing at 50Hz.

MTOW	3.448 kg
Wings	8 (W1p, W1s, W2p, W2s, W3p, W3s, W4p, W4s)
Motors	8 × GoolRC 5010-14 360KV — one per wing
ODrive boards	4 × ODrive v3.6 — axis0=port, axis1=stbd per pair
AoA sensing	MCP3208 8-ch ADC on SPI1 CE1 — pitch servo feedback
Sweep sensing	MCP3208 8-ch ADC on SPI1 CE0 — sweep servo feedback
Stroke sensing	ODrive pos_estimate — no new hardware
Wave Library	44,127 lines Python
Active hypotheses	24 (H-01 to H-27 excluding H-09/10/11)
Document version	Revision 14 — July 2026

1. AIRFRAME AND STRUCTURE

1	set	Main airframe / fuselage	Carbon fibre or composite. Mounts all 8 motor assemblies, all 16 servos, all electronics. Custom fabricated.	Custom
8	pcs	Wing panels — S1223 profile, 165mm chord	Lightweight composite or film-over-foam. Each wing is independent. Must withstand 96° amplitude at 3Hz.	Custom
8	pcs	Sector wheel assemblies	100° mechanical limit, one per wing. Converts motor rotation to flapping arc. No shared drive between any wings.	Custom
8	sets	Motor-to-sector rope drive	One rope drive per wing connecting motor drum to sector wheel. Gear ratio to suit dynamics. Inspect before every flight. NOTE: Each wing has its own independent rope drive.	Cord
8	pcs	Wing pivot hardware	Pitch servo horn-to-wing attachment. Allows full AoA range without binding.	Hardware
1	set	Landing gear	Skids for land operations. Float mounts for future water ops.	Custom

2. MOTORS — 8 INDEPENDENT ASSEMBLIES

8	pcs	GoolRC 5010-14 Brushless Motor, 360KV	12N14P, 7 pole pairs. ONE MOTOR PER WING — each wing has its own independent motor and sector wheel. Verify: 5010-14 designation. KV=360, pole pairs=7. Connected to ODrive axis: axis0=port, axis1=stbd on each board.	~$25 ea
8	pcs	Motor mount hardware	Bolts and standoffs. Thread lock on all motor mount fasteners.	Hardware
8	pcs	Motor output drum / pulley	Drum on motor shaft for rope drive to sector wheel. Sized to achieve target gear ratio.	Custom

3. ODRIVE MOTOR CONTROLLERS — 4 BOARDS, 8 AXES

4	pcs	ODrive v3.6 motor controller	Each board drives TWO independent motors: axis0 and axis1. Board 0: axis0=W1port, axis1=W1stbd. Board 1: axis0=W2port, axis1=W2stbd. Board 2: axis0=W3port, axis1=W3stbd. Board 3: axis0=W4port, axis1=W4stbd. CAN node IDs: 0,1,2,3. vel_limit=150 rad/s (set via setup script). Each axis reports independent pos_estimate for stroke sensing.	~$130 ea
1	pcs	MCP2515 SPI-to-CAN adapter	Oscillator 8MHz. Interrupt on GPIO25. Pi SPI0 CE0 → CAN bus. Connects to all 4 ODrives in chain.	~$5
2	m	CAN bus twisted-pair cable	CAN-H and CAN-L. Daisy-chains ODrives 0→1→2→3.	Per length
2	pcs	CAN bus termination resistors, 120Ω	One at each end of the CAN bus chain. Required.	<$1

4. PITCH SERVOS — ANGLE OF ATTACK, 8 WINGS

8	pcs	TORQ TCM-1001 or equiv. high-speed digital servo	333Hz PWM. 7.4V supply. MUST have analog feedback potentiometer output pin for MCP3208 ADC reading. GPIO pins (Pi PWM): W1p=18, W1s=23, W2p=24, W2s=25, W3p=12, W3s=16, W4p=20, W4s=21. Resolution: 0.1°. Controls angle of attack independently per wing — the key per-wing AoA control capability.	~$15 ea
8	wires	Pitch servo feedback signal wires	Servo feedback pot output → MCP3208 #1 ADC channel. One wire per wing. Keep under 30cm. Add 1kΩ series resistor at ADC input for protection. W1p→CH0, W1s→CH1, W2p→CH2, W2s→CH3, W3p→CH4, W3s→CH5, W4p→CH6, W4s→CH7.	<$1

5. SWEEP SERVOS — FORE-AFT POSITION, 8 WINGS

8	pcs	DS3225 or equiv. high-torque digital servo	25kg-cm, digital, 7.4V, standard 50Hz PWM. MUST have analog feedback potentiometer output pin. GPIO pins (Pi PWM): W1p=4, W1s=17, W2p=27, W2s=22, W3p=5, W3s=6, W4p=13, W4s=19.	~$10 ea
8	wires	Sweep servo feedback signal wires	Sweep servo feedback pot → MCP3208 #2 ADC channel. W1p→CH0, W1s→CH1, W2p→CH2, W2s→CH3, W3p→CH4, W3s→CH5, W4p→CH6, W4s→CH7. Add 1kΩ series resistor at ADC input.	<$1

6. AoA AND SWEEP FEEDBACK ADC SYSTEM

2	pcs	MCP3208 8-channel 12-bit SPI ADC	Microchip MCP3208-CI/P or compatible. ADC #1 (pitch feedback): SPI Bus 1, CE1. ADC #2 (sweep feedback): SPI Bus 1, CE0. Both on same SPI bus, different chip selects. VREF: 3.3V from Pi. Sample rate: all 8 ch in ~107µs. Resolution: 12-bit → ~0.044°/step at servo. Enable: dtoverlay=spi1-2cs in /boot/config.txt. Pi connections: CLK=GPIO11, MOSI=GPIO10, MISO=GPIO9, CE0=GPIO8, CE1=GPIO7.	~$3 ea
1	pcs	1kΩ resistor network (16 resistors)	Series protection on each of 16 feedback signal lines (8 pitch + 8 sweep). Prevents ADC input from short to 5V servo rail. 0.25W minimum.	<$1

7. FLIGHT COMPUTER

1	pcs	Raspberry Pi 4 Model B, 4GB RAM	4GB required — 2GB insufficient for extended discovery flights. Runs Wave Library at 50Hz with 55-layer tick stack. Mean tick: 0.90ms (dev machine), ~11ms estimated on Pi 4.	~$55
1	pcs	MicroSD card, 64GB, Class 10 or A1	Flight logs at 50Hz: ~50MB/hour. 64GB gives ~1280 flight hours. High endurance variant recommended (frequent writes).	~$10
1	pcs	Pi heatsink and case with airflow	Motor vibration environment. Must be securely mounted. Passive heatsink minimum; active fan recommended for extended summer Discovery Mode flights.	~$10

8. SENSORS

1	pcs	MPU-6050 IMU (I2C, addr 0x68)	Primary attitude reference. 6-DOF, 100Hz. Provides roll/pitch/yaw rates and accelerations. Used by stall detection Tier 1 and Tier 4.	~$3
1	pcs	BMP280 barometer (I2C, addr 0x76)	Altitude above launch point. Used above 3m where rangefinder loses terrain-following accuracy.	~$3
1	pcs	TF-Luna lidar rangefinder (50Hz)	Height above ground/water below 10m. Primary height reference for landing, approach, and water ops. Over water: reads actual wave surface height.	~$20
1	pcs	DS18B20 waterproof temperature sensor	Outside air temperature on GPIO4 (One-Wire bus). 4.7kΩ pull-up to 3.3V required. Enable: dtoverlay=w1-gpio,gpiopin=4 in /boot/config.txt. Required for density altitude correction at elevation.	~$3
1	pcs	HX711 ADC + strain gauge load cell, 50N+	TC-01 static thrust measurement. Not flown. DT=GPIO26, SCK=GPIO7. Requires mco_loadcell.py.	~$12

9. RADIO, NAVIGATION, PANEL

1	set	RC transmitter + receiver (SBUS/IBUS, 4ch min)	Ch1=Roll, Ch2=Pitch, Ch3=Lever, Ch4=Yaw. Verify channel mapping before TC-02.	Per system
1	pcs	Pixhawk flight controller	Sensor bridge only — does not run flight control. Provides GPS, compass, additional IMU via MAVLink UART at 921600 baud to Pi.	~$40
1	pcs	GPS module for Pixhawk	External GPS+compass. Mount away from motors.	~$20
1	pcs	Touchscreen display, 3.5" minimum, SPI or HDMI	Runs mco-panel.service. Shows 4 function buttons and 6 margin bars (altitude, stall, freq, amplitude, gust, combined) in yellow/amber/red.	~$25

10. POWER SYSTEM

1	pcs	Main LiPo, 4S, 4000mAh minimum, 30C	Powers 8 motors through 4 ODrives. Higher capacity for longer Discovery Mode flights.	Per spec
1	pcs	Servo LiPo, 2S, 1500mAh, 7.4V nominal	Dedicated supply for all 16 servos. Separate from main battery prevents motor noise on servo rail.	Per spec
1	pcs	BEC 5V 3A regulated	Clean Pi power supply. Motor switching noise on Pi 5V rail can cause SD card corruption.	~$8
1	pcs	30A blade fuse + holder	Main battery lead fuse.	~$3
1	pcs	5A blade fuse + holder	Pi power supply fuse.	~$2

11. COMMISSIONING TEST EQUIPMENT

1	set	Static thrust test fixture	Rigid frame for TC-01 and TC-02 thrust measurement. Must withstand full thrust (>40N) without movement. HX711 + load cell + fixture = lift-frequency curve.	Custom
1	pcs	Infrared thermometer or thermal camera	Motor temperature monitoring during TC-01-06/07 frequency ceiling test.	~$20
1	pcs	CAN bus test adapter (optional)	USB-CAN adapter for direct ODrive diagnostics.	~$30
1	pcs	Laptop for ground station	Runs mco_ground_station.py. WiFi to aircraft. Second operator required during all TC flights.	Per availability

12. CONSUMABLES AND TOOLS

1	roll	Thread lock, medium (Loctite 243 blue)	All motor mount fasteners. All sector wheel fasteners. Do not use on rope drive adjustment hardware.	Per roll
1	can	Conformal coating (for water ops)	Apply to Pi board, ODrive boards, servo electronics if water operations are planned.	Per can
16	pcs	1kΩ 0.25W resistors	8 for pitch feedback lines, 8 for sweep feedback lines. Series protection at MCP3208 ADC inputs.	<$1

13. SOFTWARE — NO COST

1	file	mco_wave_library.py	44,127 lines. Main flight control library. 24 active hypotheses (H-01 to H-27 excl H-09/10/11). 90 configuration variants. Full optimizer, sensing, fault tolerance, and discovery systems.	No cost
10	files	Supporting Python files	mco_main, mco_mavlink, mco_gs_server, mco_logger, mco_odrive_setup, mco_panel, mco_ground_station, mco_sync, mco_hypothesis_tool, mco_loadcell (pending).	No cost
1	file	Raspberry Pi OS Lite 64-bit	From raspberrypi.com. Lite (no desktop) required for maximum control loop headroom.	No cost

14. SUMMARY

TOTAL MOTORS	8 (one per wing, independent)
TOTAL ODRIVE BOARDS	4 (2 motors each = 8 axes total)
TOTAL PITCH SERVOS	8
TOTAL SWEEP SERVOS	8
TOTAL SERVOS	16
PITCH FEEDBACK ADC	1 × MCP3208, SPI1 CE1 (8 ch)
SWEEP FEEDBACK ADC	1 × MCP3208, SPI1 CE0 (8 ch)
CAN NODES	4 (ODrive boards 0-3)
PER-WING SENSING	AoA ✓ Sweep ✓ Stroke ✓ Dihedral: open-loop
WAVE LIBRARY	44,127 lines
ACTIVE HYPOTHESES	24 including 4 MCO-1D originals
TICK TIME Pi4 est.	~11ms mean (1.8× headroom vs 20ms)
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✓ NOTE: The stroke position sensing (per-wing flap position) requires no new hardware. Reading ODrive axis0.pos_estimate and axis1.pos_estimate on each of the 4 boards gives 8 independent wing stroke positions via the existing CAN bus.

✓ NOTE: Both MCP3208 ADC chips use the same SPI bus (SPI1) with different chip selects (CE0 for sweep, CE1 for pitch). Enable both with: dtoverlay=spi1-2cs in /boot/config.txt.

■ WARNING: Each wing has its own independent motor and sector wheel. There is NO shared rope drive between port and starboard on any pair. This is different from the previous architecture. All software constants have been corrected to MOTOR_COUNT=8.